

Assignment-03

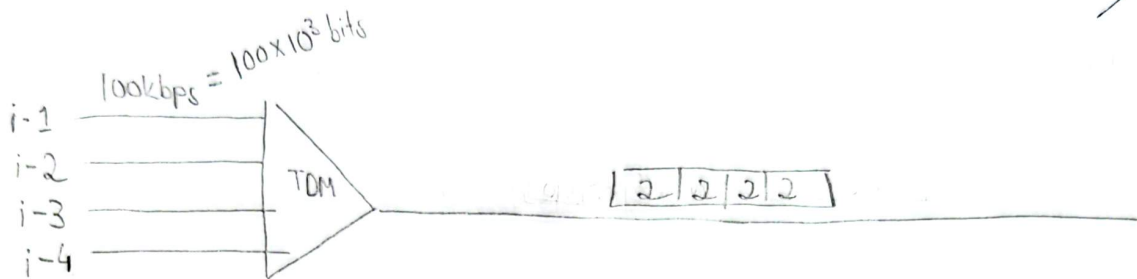
Name : Afnan Mohammad Hafiz

ID : 23241004

Sec : 13

CSE320

Ans. to the ques. no. 1



$$\text{Frame rate} = \frac{\text{Input rate}}{\text{Unit}} = \frac{100 \times 10^3}{2} = 50000 \text{ fps}$$

$$\text{Frame duration} = \frac{1}{\text{Frame rate}} = \frac{1}{50000} = \cancel{0.02 \text{ ms}} 0.02 \text{ ms}$$

$$\text{Bit rate} = \text{Frame size} \times \text{Frame rate}$$

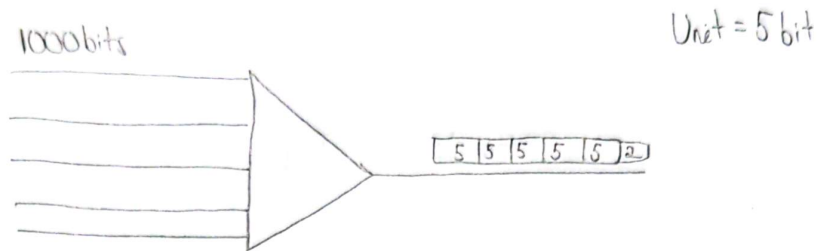
$$= 8 \times 50000$$

$$= 400 \text{ kbps}$$

$$\text{Frame size} = 4 \times 2 = 8 \text{ bit}$$

$$\text{Bit duration} = \frac{1}{400 \times 10^3} = 2.5 \mu\text{s}$$

Ans. to the ques no. 2



a) Input rate = 1000 bits

" duration = $\frac{1}{\text{Input rate}} = \frac{1}{1000} = 1 \times 10^{-3} \text{ s}$

b) Frame size = $(\text{No. of input slot} \times \text{Unit}) + \text{sync bits} = 5 \times 5 + 2 = (25+2) \text{ bits}$
~~Unit~~ = 27 bits

Frame rate = $\frac{\text{Input rate}}{\text{Unit}} = \frac{1000}{5+2} = 200 \text{ fps}$

Frame duration = $\frac{1}{\text{Frame rate}} = \frac{1}{200} = 5 \times 10^{-3} \text{ s}$

$$\begin{aligned}
 \text{c) Output rate} &= \text{Frame size} \times \text{Frame rate} \\
 &= 27 \times 200 \\
 &= 5400 \text{ bps}
 \end{aligned}$$

$$\text{Output duration} = \frac{1}{\text{output rate}} = \frac{1}{5400} = 1.851 \times 10^{-4} \text{ s}$$

$$\begin{aligned}
 \text{d) Output slot duration} &= \text{Output duration} \times \text{Unit} \\
 &= 1.851 \times 10^{-4} \times 5 \\
 &= 9.255 \times 10^{-4} \text{ s}
 \end{aligned}$$

Ans. to the ques. no. 3

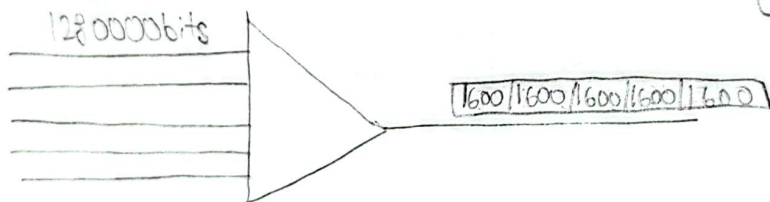
~~Characters total = $1000 \times 20 \times 40$~~

In 5 seconds pages processed = 1000

" " " " " " = $\frac{1000}{5} = 200$ pages

$\therefore$ Input rate = $200 \times 20 \times 40 \times 8$
= 1280000 bits

Unit = $5 \times 40 \times 8 = 1600$ bits



a) Input rate = 1280000 bits

" duration = $\frac{1}{1280000} = 7.8125 \times 10^{-7} s$

b) Frame size = $(5 \times 1600 + 5)$
= 8000 + 5
= 8005 bits

$$\text{Frame rate} = \frac{1280000}{1600} = 800 \text{ fps}$$

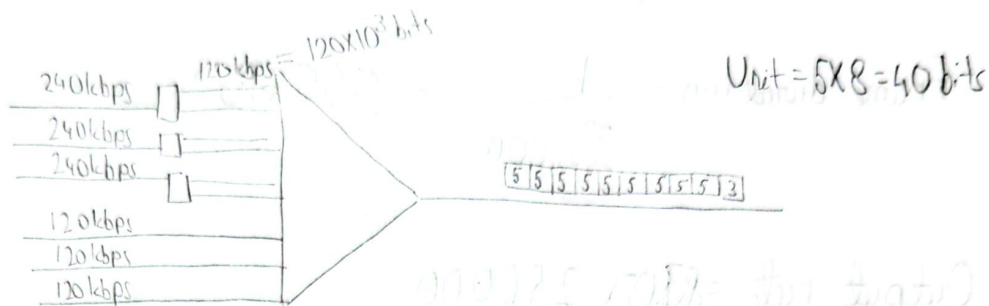
$$\text{Frame duration} = \frac{1}{800} = 1.25 \times 10^{-3} \text{ s}$$

$$\begin{aligned} \text{c) Output rate} &= 8005 \times 80000 \\ &= 64040000 \text{ bps} \end{aligned}$$

$$\text{Output duration} = \frac{1}{64040000} = 1.561 \times 10^{-7} \text{ s}$$

Ans. to the ques. no. 4

a)



b) Input rate = $120 \text{ kbps} = 120 \times 10^3 \text{ bits}$

" duration = $\frac{1}{120 \times 10^3} = 8.33 \times 10^{-6} \text{ s}$

c) Frame size = $9 \times 40 = 363 \text{ bits}$

" rate = $\frac{120 \times 10^3}{40} = 3000 \text{ fps}$

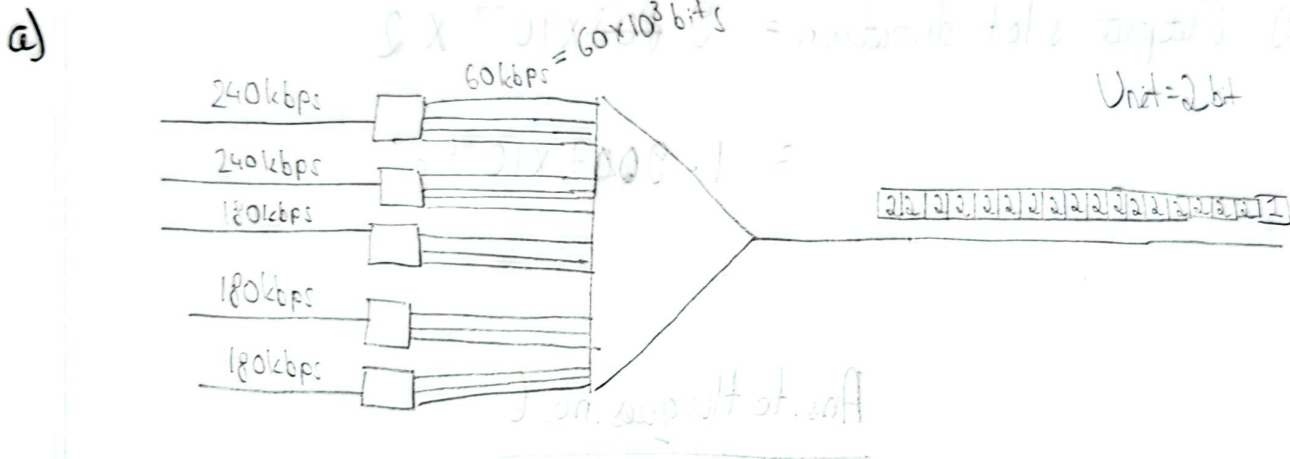
" duration = $\frac{1}{3000} = 3.33 \times 10^{-4} \text{ s}$

d) Output rate = 363×3000
 $= 1089 \times 10^3 \text{ bps}$

" duration = $\frac{1}{1089 \times 10^3} = 3.67 \times 10^{-5} \text{ s}$

e) Output slot duration = $9.183 \times 10^{-7} \times 40$
 $= 3.67 \times 10^{-5} s$

Ans. to the ques. no. 5



b) Input rate = $60 \text{ kbps} = 60 \times 10^3 \text{ bits/s}$

" duration = $\frac{1}{60 \times 10^3} = 1.67 \times 10^{-5} s$

c) Frame size = $(17 \times 2) + 1 = 35 \text{ bits}$

" rate = $\frac{60 \times 10^3}{2} = 30000 \text{ fps}$

" duration = $\frac{1}{30000} = 3.33 \times 10^{-5} s$

$$d) \text{ Output rate} = 35 \times 30000 \\ = 1050000 \text{ bps}$$

$$" \text{ duration} = \frac{1}{1050000} = 9.523 \times 10^{-7} \text{ s}$$

$$e) \text{ Output slot duration} = 9.523 \times 10^{-7} \times 2 \\ = 1.9047 \times 10^{-6} \text{ s}$$

Ans. to the ques. no. 6

a) Statistical TDM should be used here. In this TDM, the slots allocated would not be wasted and only devices that actually have data will be allocated. The frame would shrink to fit only devices that are active.

b)

	Second slot	First slot
A	A	A
B	B	
C		C
D	D	
E	E	E
F		F
G	G	
H		H



$$j = i - 3 = d - n = 10$$

c)

	2 nd slot	1 st slot
A	A	A
B	B	
C		C
D	D	
E	E	E
F		F
G	G	
H		H



$$A = (1100110, 1110000)$$

$$B = (00101101, 11110000)$$

$$C = (10011101, 11100000)$$

$$D = (00101101, 1101010)$$

$$E = (10011101, 1100000)$$

Ans. to the ques. no. 7

a) Here

$$n = 8$$

$$k = 2$$

We know,

$$n = k + r$$

$$\therefore r = n - k = 8 - 2 = 6$$

b) Total possible codewords = $2^n = 2^8 = 256$

" " datawords = $2^k = 2^2 = 4$

" valid code words = 4

" invalid " = $256 - 4 = 252$

c) Hamming distance -

$$(00001111, 01010011) = 4$$

$$(00001111, 10110100) = 6$$

$$(00001111, 1111001) = 6$$

$$(01010011, 10110100) = 6$$

$$(01010011, 11110011) = 4$$

$$(10110100, 11111001) = 4$$

$$\therefore \text{Minimum HD} = 4$$

$$\text{Error detecting capacity } d = HD - 1 = 4 - 1 = 3$$

$$\text{" correcting } \therefore t = \left\lfloor \frac{HD-1}{2} \right\rfloor = \left\lfloor \frac{4-1}{2} \right\rfloor = \left\lfloor \frac{3}{2} \right\rfloor = \lfloor 1.5 \rfloor = 1$$

1) Hamming distance with 1st -

$$(00011101, 00001111) = 2$$

$$(00011101, 01010011) = 4$$

$$(00011101, 10110100) = 4$$

$$(00011101, 11111001) = 4$$

$$\text{Minimum HD} = 2 \geq 1 \text{ (Correcting capacity)}$$

$\therefore$ It is corrupted and cannot be corrected.

Hamming distance with 2nd -

$$(10111100, 00001111) = 5$$

$$(10111100, 01010011) = 7$$

$$(10111100, 10110100) = 1$$

$$(10111100, 11111001) = 3$$

Minimum HD = 1 $\geq$ 1 (correcting capacity)

$\therefore$ It is corrupted and can be corrected

Codeword = 10110100

$\therefore$ Dataword = 10

Ans. to the ques. no. 8

a) $X = 1004$

$$D = 1004 \div 256 = 236 = 11101100$$

$$G = 1004 \div 16 = 12 = 1100$$

Here,

$$\begin{array}{r|l}
 1100 & 11101100 \\
 \hline
 & 1100 \\
 \hline
 & 001011 \\
 & 1100 \\
 \hline
 & 01110 \\
 & 1100 \\
 \hline
 & 00100
 \end{array}$$

That

Here,

$$n = 4$$

$$n-1 = 3$$

Augmented Codeword -

$$\begin{array}{r}
 \cancel{11101000} \\
 11101100 \quad 000
 \end{array}$$

That is,

$$\begin{array}{r}
 1100 \overline{) 11100100100} \\
 \underline{1100} \\
 001001 \\
 \underline{1100} \\
 01010 \\
 \underline{1100} \\
 01100 \\
 \underline{1100} \\
 0000100
 \end{array}$$

Here, remainder $\neq 0$. Error can be detected.

Ans. to the ques. no. 9

a) Generator = 4 bit

$$\therefore n-1 = 4-1 = 3 \text{ bit}$$

Codeword = 9 bits

$$\therefore \text{Dataword} = 9-3 = 6 \text{ bits}$$

Thus,

$$C(n, k) = C(9, 6)$$

We know,

$$n = k + r$$

$$\therefore r = n - k = 9 - 6 = 3 \text{ bits}$$

b) Total possible codewords = $2^n = 2^9 = 512$

" " dataword = $2^k = 2^6 = 64$

" valid codewords = $2^k = 64$

" invalid " = $512 - 64$
 $= 448$

c) $ID = \text{Even}$

1st codeword = 100110100

$= x^8 + x^5 + x^4 + x^2$

2nd Codeword = 100110010

$= x^8 + x^5 + x^4 + x$

1st Generator = 1001

$= x^3 + 1$

2nd Generator = 1001

$= x^3 + 1$

First -

$$\begin{array}{r|l}
 x^3+1 & x^8+x^5+x^4+x^2 \\
 \underline{x^8+x^5} & \\
 & x^4+x^2 \\
 & \underline{x^4+x} \\
 & x
 \end{array}
 \quad
 \begin{array}{l}
 \frac{x^8}{x^3} = x^5 \\
 \frac{x^4}{x^3} = x
 \end{array}$$

Here, remainder $\neq 0$. So it is corrupted

~~2nd~~ Second -

$$\begin{array}{r|l}
 x^3+1 & x^8+x^5+x^4+x \\
 \underline{x^8+x^5} & \\
 & x^4+x \\
 & \underline{x^4+x} \\
 & 0
 \end{array}
 \quad
 \begin{array}{l}
 \frac{x^8}{x^3} = x^5 \\
 \frac{x^4}{x^3} = x
 \end{array}$$

Here, remainder = 0. So it is corrected.

Ans. to the ques. no. 10

$$a) (4501)_{16} = (1000101000000001)_2$$

$$(1A4F)_{16} = (1101001001111)_2$$

$$(A309)_{16} = (1010001100001001)_2$$

$$+ (B007)_{16} = (1011000000000111)_2$$

$$\begin{array}{r} \text{00018260} \\ \hline (10110010010011000000)_2 \end{array}$$

$$(1011001001100001)_2$$

Now,

$$(1011001001100001)_2 = (B261)_{16} \quad [\text{Wrap Around Sum}]$$

Now, 1st complement -

$$(0100110110011110)_2 = (4D9E)_{16} \quad [\text{Check Sum}]$$

$$b) 2^{\text{nd}} \text{ word} = (1A4F)_{16} = (1A4A)_{16}$$

Adding all received words -

4501

1A4A

A309

8007

$$\begin{array}{r} 4501 \\ 1A4A \\ A309 \\ 8007 \\ \hline 409E \\ \hline 1FFF9 \end{array}$$

$$\therefore 1FFF9 = 1111111111111001$$

$$\begin{array}{r} 1 \\ \hline (1111111111111010)_2 \end{array} \quad [\text{Wrapped Around Sum}]$$

1's complement -

$$(0000000000000101)_2 = 5$$

Here,

5 $\neq$ 0, thus error is detected.

Ans. to the ques. no. 11

a) The figure refers to CSMA/CD.

$$\begin{aligned}\text{b) Transmission Time, } T_{fr} &= t_5 - t_1 \\ &= 5\mu s - 1\mu s \\ &= 4\mu s\end{aligned}$$

$$\begin{aligned}\text{c) Maximum time required, } T_p &= t_4 - t_1 \\ &= 2.8\mu s - 1\mu s \\ &= 1.8\mu s\end{aligned}$$

d) We know in CSMA/CD -

$$\begin{aligned}\text{Vulnerable Time} &= T_p \\ &= 1.8\mu s\end{aligned}$$

In pure aloha -

$$\text{Vulnerable Time} = 2 T_{fr} = 2 \times 4 = 8\mu s$$

In slotted aloha -

$$\text{Vulnerable Time} = T_{fr} = 4\mu s$$

$\therefore$ CSMA/CD < Slotted Aloha < Pure Aloha

CSMA/CD has the lowest vulnerable time because the station sense the channel first and wait until there is no data transmission as to avoid collision.

e) $v = 10 \text{ Mbps}$
 $= 10 \times 10^6 \text{ bps}$

We know,

$$T_{fr} \geq 2T_p$$

Now,

$$s = vt$$

$$= 10 \times 10^6 \times 2 \times 1.8 \times 10^{-6}$$

$$= 36 \text{ bits}$$

Ans. to the ques. no. 12

a) 0-4 = Idle

4-10 = Transmission

10-14 = Collision

14-20 = Idle

20-25 = Transmission

25-30 = Collision

30-40 = Idle

b) Total transmission time = $(4-10) + (20-25)$
 $= 6 + 5$

$= 11 \text{ seconds}$

Total collision time = $(10-14) + (25-30)$
 $= 4 + 5$

$= 9 \text{ seconds}$

c) If station want to send a frame, it should be in idle time which is in -

$(0-4)s, (14-20)s, (30-40)s$

Ans. to the ques. no. 13

a) We know

$$s = vt$$

$$\text{or, } t = \frac{s}{v} = \frac{1200 \times 8}{30 \times 10^6} = 3.2 \times 10^{-4} s$$

b) We know,

$$\begin{aligned} T_{fr} &\geq 2T_p \\ T_{fr} &\geq 2 \times 5 \times 10^{-4} \end{aligned}$$

$$T_{fr} = 2T_p$$

$$\text{or, } 3.2 \times 10^{-4} = 2T_p$$

$$\text{or, } T_p = \frac{3.2 \times 10^{-4}}{2}$$

$$= 1.6 \times 10^{-4} s$$

$$\begin{aligned} \therefore \text{Vulnerable time} &= T_p \\ &= 1.6 \times 10^{-4} s \end{aligned}$$

c) We know,

$$\text{Distance} = T_p \times v$$

$$= 1.6 \times 10^{-4} \times 5 \times 10^7$$

$$= 8000 \text{ m}$$

Ans. to the ques. no. 14

a) ~~Non-persistent~~ Non-persistent CSMA

b) In non-persistent CSMA, there can be increased delay. When the channel is busy, it waits for a while before checking it again. During this wait time, the channel might be free without the station sensing it, which then leads to unnecessary delay and reduced efficiency.

c) CSMA/CD can detect collision. After the collision is detected, the station waits for a while until the channel voltage become 0, and then ~~sends~~ starts transmission. If collision is detected again, the station backoff time doubles, and it repeats upto a maximum number of attempts.